C2_sep02

miércoles, 2 de septiembre de 2026 9:17

Ejemplo

Una muestra aleatoria de n observaciones, $Y_1, Y_2, \ldots, Y_n$ se selecciona de una población en la que Y_i, para $i = 1, 2, \ldots, n$, posee una función de densidad de probabilidad uniforme en el intervalo $(0, \theta)$ donde θ es desconocida. Use el método de momentos para estimar el parámetro θ.

$$y_i \sim U(0,\theta)$$

si $k=1$

$$\mu_1' = E(Y) = \frac{0+\theta}{2} = \frac{\theta}{2}$$

$$m_1' = \frac{1}{n}\sum y_i = \bar{y}$$

Ahora igualamos.

$$\mu_1' = m_1'$$

$$\frac{\theta}{2} = \bar{y}$$

$$\boxed{\hat{\theta} = 2\bar{y}}$$

Como debemos estimar dos parámetros debemos igualar dos momentos.

$$y_i \sim \text{Gamma}(\alpha,\beta)$$

$$\mu = E(Y) = \alpha\beta; \quad \sigma^2 = Var(Y) = \alpha\beta^2 =$$

$$\mu_1' = E(Y) = \alpha\beta \qquad m_1' = \bar{y}$$

$$\mu_2' = E(Y^2) \qquad m_2' = \frac{1}{n}\sum y_i^2$$

$$\mu_1' = m_1' \qquad\qquad \mu_2' = m_2'$$

$$① \quad \alpha\beta = \bar{y} \qquad\qquad E(y^2) = \frac{1}{n}\sum y_i^2$$

Sabemos que

$$\sigma^2 = E(y^2) - E(Y)^2$$

$$\alpha\beta^2 = E(y^2) - (\alpha\beta)^2$$

$$E(y^2) = \alpha\beta^2 + \alpha^2\beta^2$$

$$② \quad * \quad E(y^2) = \alpha\beta^2 + \alpha^2\beta^2 = \frac{1}{n}\sum y_i^2$$

De la ec. ① despejamos β:

$$\hat{\beta} = \frac{\bar{y}}{\hat{\alpha}}$$

y vamos a reemplazar en la ec ②

$$\hat{\alpha}\left(\frac{\bar{y}}{\hat{\alpha}}\right)^2 + \hat{\alpha}^2\left(\frac{\bar{y}}{\hat{\alpha}}\right)^2 = \frac{1}{n}\sum y_i^2$$

$$\frac{\bar{y}^2}{\hat{\alpha}} + \bar{y}^2 = \frac{1}{n}\sum y_i^2$$

$$\frac{\bar{y}^2}{\hat{\alpha}} = \frac{1}{n}\sum y_i^2 - \bar{y}^2$$

$$\frac{\bar{y}^2}{\frac{1}{n}\sum y_i^2 - \bar{y}^2} = \hat{\alpha} \Rightarrow \frac{\bar{y}^2}{\frac{\sum y_i - n\bar{y}^2}{n}} = \frac{n\bar{y}^2}{\sum y_i - n\bar{y}^2}$$

$$\boxed{\frac{n\bar{y}^2}{\sum(y_i-\bar{y})^2} = \hat{\alpha}}$$

$$\hat{\beta} = \frac{\bar{y}}{\hat{\alpha}}$$

$$= \frac{\bar{y}}{\frac{n\bar{y}^2}{\sum(y_i-\bar{y})^2}}$$

$$= \frac{\bar{y}\sum(y_i-\bar{y})^2}{n\bar{y}^2}$$

$$\boxed{\hat{\beta} = \frac{\sum(y_i-\bar{y})^2}{n\bar{y}}}$$

$$L(\theta) = L(p) = L(y_1,\ldots,y_n \mid p)$$

$$= p(y_1 \mid p)\,p(y_2 \mid p)\cdots p(y_n \mid p)$$

$$= p^{y_1}(1-p)^{1-y_1}\; p^{y_2}(1-p)^{1-y_2} \times \cdots \times p^{y_n}(1-p)^{1-y_n}$$

$$= p^{y_1+y_2+\ldots+y_n}\,(1-p)^{1-y_1+1-y_2+\ldots+1-y_n}$$

$$= p^{\sum y_i}\,(1-p)^{\sum(1-y_i)}$$

$$= p^{\sum y_i}\,(1-p)^{n-\sum y_i} \quad ; \text{ tomemos } y = \sum y_i$$

$$= p^y(1-p)^{n-y}$$

Ahora calculamos el logaritmo

$$\ln(L(p)) = \ln\left(p^y(1-p)^{n-y}\right)$$

$$= \ln p^y + \ln(1-p)^{n-y}$$

$$= y\ln p + (n-y)\ln(1-p)$$

$$\frac{\partial \ln(L(p))}{\partial p} = \frac{y}{p} - \frac{(n-y)}{1-p}$$

Ahora igualamos a 0 y desp~~e~~ajamos:

$$\frac{y}{\hat{p}} - \frac{(n-y)}{1-\hat{p}} = 0$$

$$\frac{y}{\hat{p}} = \frac{(n-y)}{1-\hat{p}}$$

$$y(1-\hat{p}) = \hat{p}(n-y)$$

$$y - \cancel{y\hat{p}} = n\hat{p} - \cancel{y\hat{p}}$$

$$y = n\hat{p}$$

$$\boxed{\frac{y}{n} = \hat{p}}$$

$$f(y_i \mid \mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{1}{2\sigma^2}(y_i-\mu)^2\right\}$$

$$L(\mu,\sigma^2) = L(y_1,\ldots,y_n \mid \mu,\sigma^2)$$

$$= f(y_1 \mid \mu,\sigma^2)\,f(y_2 \mid \mu,\sigma^2)\cdots f(y_n \mid \mu,\sigma^2)$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{1}{2\sigma^2}(y_1-\mu)^2\right\} \times \cdots \times \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{1}{2\sigma^2}(y_n-\mu)^2\right\}$$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left\{-\frac{1}{2\sigma^2}(y_1-\mu)^2 - \frac{1}{2\sigma^2}(y_2-\mu)^2 - \cdots - \frac{1}{2\sigma^2}(y_n-\mu)^2\right\}$$

$$= \left(\frac{1}{2\pi\sigma^2}\right)^{n/2}\exp\left\{-\frac{1}{2\sigma^2}\sum(y_i-\mu)^2\right\}$$

$$\ln(L(\mu,\sigma^2)) = \ln\left(\left(\frac{1}{2\pi\sigma^2}\right)^{n/2}\exp\left\{-\frac{1}{2\sigma^2}\sum(y_i-\mu)^2\right\}\right)$$

$$= \ln\left(\frac{1}{2\pi\sigma^2}\right)^{n/2} - \frac{1}{2\sigma^2}\sum(y_i-\mu)^2$$

$$\frac{\partial \ln(L(\mu,\sigma^2))}{\partial \mu} = 0 + \frac{1}{\sigma^2}\sum(y_i-\mu)$$

Ahora igualamos a 0.

$$\frac{1}{\hat{\sigma}^2}\sum(y_i-\hat{\mu}) = 0$$

$$\sum(y_i-\hat{\mu}) = 0$$

$$\sum y_i - \sum\hat{\mu} = 0$$

$$\sum y_i - n\hat{\mu} = 0$$

$$\hat{\mu} = \frac{\sum y_i}{n}$$

$$\boxed{\hat{\mu} = \bar{y}}$$

$$\ln(L(\mu,\sigma^2)) = \ln\left(\frac{1}{2\pi\sigma^2}\right)^{n/2} - \frac{1}{2\sigma^2}\sum(y_i-\mu)^2$$

$$= \frac{n}{2}\ln\left(\frac{1}{2\pi\sigma^2}\right) - \frac{1}{2\sigma^2}\sum(y_i-\mu)^2$$

$$= \frac{n}{2}\left(-\ln(2\pi\sigma^2)\right) - \frac{1}{2\sigma^2}\sum(y_i-\mu)^2$$

$$= -\frac{n}{2}\ln(\sigma^2) - \frac{n}{2}\ln(2\pi) - \frac{1}{2\sigma^2}\sum(y_i-\mu)^2$$

$$\frac{\partial \ln(L(\mu,\sigma^2))}{\partial \sigma^2} = \left(-\frac{n}{2}\right)\left(\frac{1}{\sigma^2}\right) + \frac{2}{4\sigma^4}\sum(y_i-\mu)^2$$

$$= -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum(y_i-\mu)^2$$

Igualamos a 0

$$-\frac{n}{2\hat{\sigma}^2} + \frac{1}{2\hat{\sigma}^4}\sum(y_i-\hat{\mu})^2 = 0$$

$$\frac{n}{2\hat{\sigma}^2} = \frac{1}{2\hat{\sigma}^4}\sum(y_i-\hat{\mu})^2$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum(y_i-\hat{\mu})^2$$

$$\boxed{\hat{\sigma}^2 = \frac{1}{n}\sum(y_i-\bar{y})^2}$$